

SATHISH KUMAR

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ABOUT ME

- Dedicated to unraveling complexities and empowering others.
- As a skilled debugger with a passion for understanding code, I thrive on solving technical challenges.
- My commitment to continuous learning drives **me** to excel in my projects and career.
- With a knack for breaking down problems and providing effective solutions, I'm a valuable asset to any team.

PROFESSIONAL SUMMARY

- Ability to architect, implement, and test complex embedded systems telematics framework across various platforms.
- With a strong foundation in firmware, hardware, and software design, I am adept at driving projects from concept to commercialization.
- Experience in test-driven development, manufacturing processes, technical documentation, and delivering high-quality, high-volume products ensures successful project execution.

SKILLS

- C++ (11/14/17)
- Python
- Protocol Buffers
- Gtest
- RTOS
- Multi-Threading
- OOPS
- Architecture (x86, arm)
- ZeroMQ
- Modbus
- Jenkins
- Boost
- Thrift
- Linux
- gRPC
- Telematics
- Static analyzers(cppcheck, flaw finder)
- Memory analyzers(GDB,Valgrind)
- Tools (Wireshark, Canalyzer, VSCode)
- Protocols (TCP, UDP, CAN/J1939, SFTP, UART)
- Git
- Exception Handling
- Docker
- STL containers

EXPERIENCE

2021 - Present

Embedded Software Engineer | TATA Elxsi | Chennai

Project: Telematics

- Application layer development for telematics domain for off-road vehicle using C++ on Linux platform (LXC container).
- This project contains various environment with different architecture to provide solutions for the customer needs.

Roles and Responsibilities:

- Developing and maintaining software for modules like **MineStar**, **HPGPS**, **reports**, **schedulers**, **webpage** communication using thrift.
- Understanding **customer requirements** and providing solutions.
- Encoding, Decoding of Google Protobuf messages which act as **storage** medium.
- Verifying the code implemented by **unit testing** using Gtest, **static analysis** tools like flaw finder cppcheck, creating test plan and performing **functional testing**.
- Creating applications on Linux platform and integrated to the workspace so that some Realtime data can be simulated which will be helpful for debugging (**Memory analyzer**)
- Evaluating and **debugging bugs**, system related issues
- Creating **automation scripts using python** to test few features which reduce some manual efforts.

EDUCATION

2017 - 2021

Bachelor Of Engineering | Computer Science | Sri Muthukumaran Institute of Technology

- CGPA: 7.96

2015 - 2017

Higher Secondary Education | Computer Science & Mathematics | Lamech Higher Secondary School

- CGPA: 8.64

EXTRACURRICULAR ACTIVITIES

- Workshop: KaaShiv InfoTech (IoT)
- Problem Solving – [LeetCode](#)
- Cybersecurity – HackTheBox
- Sports – Badminton, Cricket, Volleyball